

Descriptor: *BLDC Hall Sensor Displacement Dataset (BLDC-HSD)*

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ABSTRACT Brushless dc (BLDC) motors depend on accurate rotor position detection via Hall sensors for optimal performance. Faults, such as sensor displacement, can disrupt commutation and lead to efficiency losses. Any research that utilizes deep learning to detect Hall sensor faults will benefit from using the BLDC-HSD dataset for training and testing their AI. BLDC-HSD was meticulously prepared and designed for this purpose. BLDC-HSD consists of phase current measurements under various Hall sensor displacement conditions, categorized as no delay, 0.0001 delay, 0.005 delay, and 0.01 delay. Each condition includes 60 000 data points recorded at intervals of 500 ns. Data are structured in an Excel file with columns for time and phase currents. This well-organized dataset supports the development of deep learning models for accurate fault detection and classification, contributing to enhanced motor control and diagnostic capabilities.

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DATA TYPE/LOCATION Image, Time-series; n/a

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INDEX TERMS Brushless dc (BLDC), brushless motors, displacement, fault detection, hall sensor.

BACKGROUND

Brushless dc (BLDC) motors have emerged as a pivotal technology in industrial applications due to their superior efficiency, low maintenance requirements, and precise control capabilities [1], [2], [3], [4]. These motors rely on accurate rotor position detection to ensure optimal performance and reliability. Hall sensors are typically employed for this purpose, providing essential data for determining precise commutation timing [5], [6], [7]. The accurate detection of rotor position allows for the precise control of the commutation process, which is crucial for maintaining the motor's efficiency and minimizing energy losses [8], [9]. For these reasons, the BLDC-HSD dataset was released which was originally used in [1]. Deep learning AI for Hall sensor fault detection can use BLDC-HSD for training and testing.

Despite their advantages, BLDC motors are not immune to issues that can compromise their performance. One significant challenge is the potential for Hall sensor faults,

particularly displacement of the sensor mount on the stator. Such displacement can lead to inaccurate signal readings, disrupting the commutation timing and adversely affecting motor operation. The resulting issues include increased energy consumption, excessive vibration, and reduced efficiency [10], [11], [12]. In severe cases, these faults can escalate to complete motor failure, posing risks of damage to both the motor and its associated components.

Addressing these challenges requires effective fault diagnosis methods that can accurately detect and classify faults related to Hall sensor displacement. Traditional diagnostic techniques may not adequately handle the complexity of these faults, which is why recent advancements in deep learning have shown promise. Deep learning, with its ability to analyze intricate patterns and features within large datasets, offers a transformative approach to fault diagnosis. This study [1] explored the application of deep learning techniques to detect Hall sensor displacement faults in BLDC

motors, aiming to improve the precision and reliability of fault diagnosis, and as a result, the BLDC-HSD dataset has been released to benefit other researchers.

COLLECTION METHODS AND DESIGN

A crucial aspect of this study [1] is the development and utilization of a well-structured dataset specifically designed for Hall sensor displacement detection. The objective of the data format and description section is to establish a comprehensive study for organizing and preparing this dataset, which includes the following.

- 1) *Defining the Dataset Structure*: Clearly outlining how the dataset is organized and categorized based on different levels of Hall sensor displacement. The dataset is divided into categories such as *no delay*, *0.0001 delay*, *0.005 delay*, and *0.01 delay*, each representing distinct fault conditions.
- 2) *Describing Data Collection and Measurement*: Providing a detailed account of the data collection process, including both real-world experiments and MATLAB simulations. This section explains how phase currents are recorded over time and how these measurements reflect the impact of Hall sensor displacement on motor performance.
- 3) *Explaining Data Format and Preprocessing*: Detailing the format of the dataset, including its structure in formats such as Excel and other suitable file types. It also covers the preprocessing steps applied to the data, such as normalization, data splitting, and augmentation, to ensure the dataset is ready for deep learning model training.
- 4) *Facilitating Model Training and Evaluation*: Ensuring that the dataset format supports effective training and evaluation of deep learning models. This involves transforming time-series data into formats suitable for input into neural networks, such as RGB images, and ensuring that the dataset is balanced and representative of various fault conditions.
- 5) *Supporting Fault Detection Analysis*: Providing a well-organized dataset that enables detailed analysis of how Hall sensor displacement affects BLDC motor performance. This includes facilitating accurate fault detection and classification, which is critical for developing reliable fault diagnosis models.

By achieving these objectives, the data format and description will play a vital role in the development of deep learning models for detecting Hall sensor displacement. The BLDC-HSD dataset aims to enhance the accuracy and reliability of fault detection in BLDC motors, ultimately contributing to more effective motor control systems and advancing the field of motor fault diagnosis. This comprehensive set of data and the insights gained from its analysis will offer valuable contributions to the optimization of BLDC motor performance and the development of robust diagnostic tools.

Raw Data Collection

The data collection process for the BLDC-HSD dataset comprises real-world experiments and MATLAB simulations carried out on BLDC motor drives under diverse operating conditions, ensuring the acquisition of a comprehensive and representative dataset. However, it is noteworthy that the dataset utilized for training and testing purposes was obtained through MATLAB simulations, following validation to ensure alignment with experimental results, chosen for ease of operation and convenience.

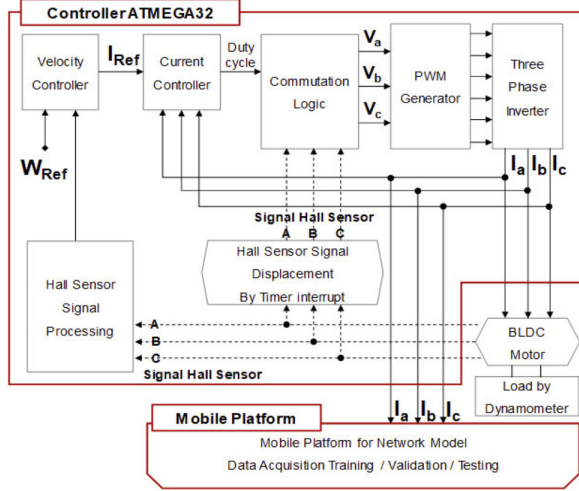
Fig. 1(a) illustrates the system setup for BLDC motor control, comprising a motor controller, commutation logic, a mobile platform for network models, and three half-bridge circuits (three-phase inverter), which are responsible for connecting the motor phases to either VCC or GND. This configuration enables the injection of current into the coils, generating the necessary magnetic fields for the different phases. A motor controller governs commutation, facilitated by the motor driver, which allows communication between the control block and the half-bridge circuits. This setup enables precise control of commutation through the circuits based on feedback from Hall position sensors, which provide information on the rotor's position to the motor controller.

To implement Hall sensor displacement or fault signals, we refined the commutation timing within the motor drive and control programming. Rather than physically adjusting the Hall sensor's position, we opted for a software-based approach.

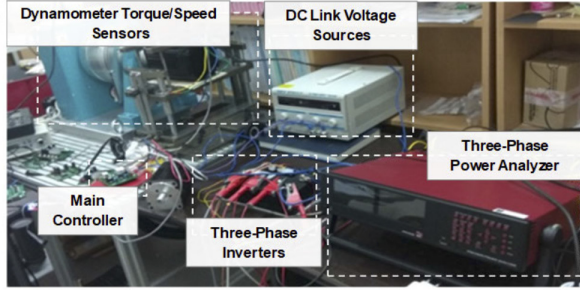
Here's how it works: After reading the hall sensor state in the main loop, a delay function is invoked to introduce the necessary delay before commutating the motor. Within this delay function, we reset a timer/counter (TCNxT) and clear its compare match flag (OCFxA). This ensures that the program waits until the TCNxT compare match flag is set, indicating that the desired delay has been reached. The commutation function is embedded within the interrupt service routine (ISR) for the TCNxT. This arrangement guarantees that the delay is executed before commutating the motor.

By employing this methodology, we achieve precise control over the commutation timing without the need for physical adjustments to the Hall sensor position. Detailed specifications of the motor drive and BLDC motor are presented in Table I.

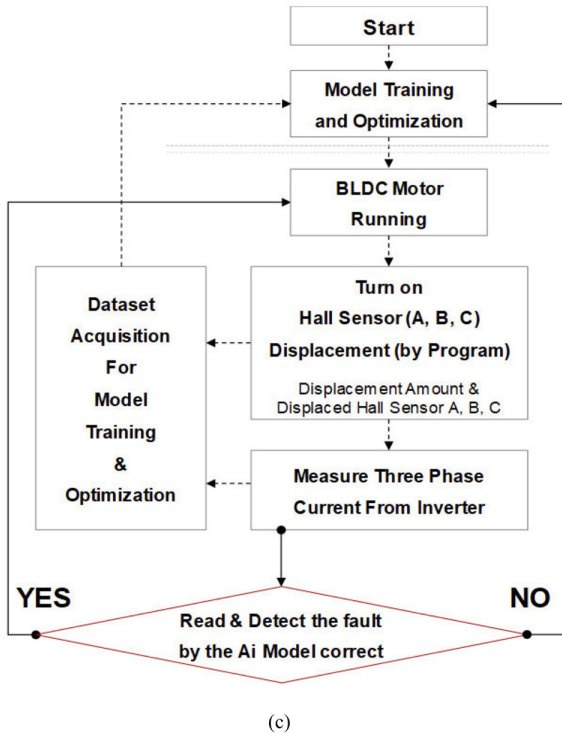
In Fig. 1(b) and 1(c), the experimental setup for driving the three-phase BLDC motor is showcased, along with an overarching flowchart depicting the diagnostic process. This setup utilizes a hysteresis brake dynamometer (BHD-144) by Validmagnetics, alongside a microcontroller for torque and rotor speed control. Additionally, a power analyzer (PPA-3500) by Newtons4th was employed for phase current and voltage measurement. Experimental results served solely for validation against MATLAB simulations, ensuring consistency between the two sets of data.



(a)



(b)



(c)

FIG. 1. Data acquisition system used and proposed system for training and testing from [1] which BLDC-HSD is based from, where (a) detailed description of the setup; (b) photo for the experimental setup; and (c) overarching flowchart depicting the diagnostic process.

TABLE I. Motor Drive and BLDC Motor Specification

Motor Drive Specification	
Controller	Atmega32 (8-bit uC)
System clock	16 MHz
PWM frequency	20 kHz
On-duty cycle	5%–90%
Motor speed range	200–2000 rpm
Load torque	2–5 N-m
DC voltage	400 V
MOSFET on-resistance	0.1 Ω
Commutation scheme	Six-step

VALIDATION AND QUALITY

Measurement Precision

The minimum pulse displacement for a 20 kHz pulse width modulation (PWM) pulse frequency is 0.0001 s, which corresponds to a displacement of two PWM pulse periods. In this study, the PWM pulse frequency is maintained at 20 kHz, and the displacement will be gradually increased until it reaches 0.01 s. At this point, the detection accuracy exceeds 99% and can be verified by monitoring the waveform itself. It is important to note that this displacement cannot be continuous, as it is governed by the PWM control scheme.

Data Processing

The measured three-phase current undergoes a meticulous preprocessing phase to create suitable input data for the neural network models. Before feeding the data into the models, the phase current is normalized to ensure uniformity and to prevent any model biases toward specific features. Normalization scales the data to a common range, typically between 0 and 1, making it easier for the models to converge during training. BLDC-HSD is divided into training and testing sets using a 4:1 ratio.

The original data are structured as a matrix of size 14700×3 , as depicted in Fig. 2(c), with each row representing phase current at 14700 different sampling times for each phase. To facilitate effective spatial feature extraction, the data are then transformed into RGB images of size $224 \times 224 \times 3$ as illustrated in Fig. 2(a) and 2(b), resulting in a 3-D input shape, exploiting its ability to detect intricate spatial patterns that may reveal valuable information about Hall sensor inaccuracies. To enhance to learn spatial patterns effectively, data augmentation techniques are applied. Augmentation involves random transformations, such as rotations, flips, and shifts, to the RGB images. This process creates additional training samples, increasing the model's exposure to diverse patterns and improving its robustness [13]. For the case where there might be an imbalance in the data distribution, balancing techniques are applied to ensure equal representation of different classes (e.g., normal and faulty Hall sensor readings). This balanced distribution prevents the models from being biased toward the majority class

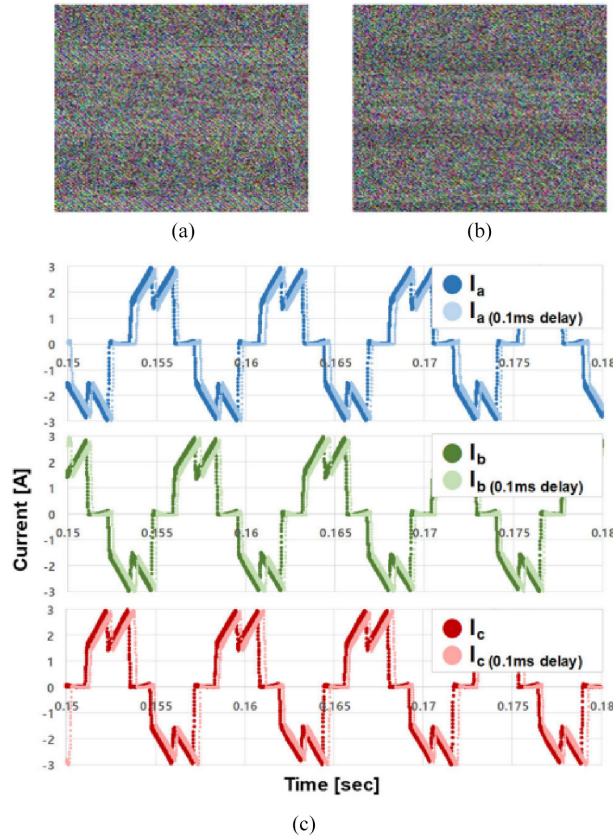


FIG. 2. (a) Three-phase current waveform in RGB without displacement in Hall sensor. (b) Three-phase current waveform in RGB with 0.1 ms displacement in Hall sensor. (c) Three-phase current waveforms with zero delay and 0.1 ms displacement [1].

and ensures fair evaluation of their accuracy in detecting Hall sensor inaccuracies.

By preprocessing the data through normalization, splitting, augmentation, sequence formation, padding, and dataset balancing, we create well-structured and representative input data [13], [14]. These carefully prepared datasets, tailored to each model's requirements, will empower the deep learning models to effectively learn from the data, improving their performance in detecting faulty Hall sensors in BLDC motor drives.

Quality for Training and Testing

BLDC-HSD for training and testing comprises three-phase currents over time, chosen because they reflect Hall sensor displacement. Commutation timing for three-phase currents is determined by the combination of three Hall sensors, as outlined in Table II. This choice is reinforced by the fact that three-phase currents are already utilized for motor speed and torque control, incurring no additional cost.

BLDC-HSD includes both normal and flawed three-phase currents, generated by introducing proper and improper commutation timing due to Hall sensor displacement at four distinct levels: 0 s (zero PWM pulse period displacement), 0.0001 s (two PWM pulse period displacement), 0.005 s

TABLE II. Six-Step Commutation Sequence for a BLDC Motor

Step	Phase A	Phase B	Phase C
1	High	Low	Open
2	High	Open	Low
3	Open	High	Low
4	Low	High	Open
5	Low	Open	High
6	Open	Low	High

(100 PWM pulse period displacement), and 0.01 s (200 PWM pulse period displacement) in A, B, and C Hall sensors. In this BLDC-HSD dataset, the PWM pulse frequency is 20 kHz. The speed/torque control of BLDC motor is achieved by varying the duty cycles of PWM Pulse from the micro-controller. BLDC-HSD is prepared to assess the detection capability at different displacement levels in Hall sensors. The BLDC motor is driven by a six-step commutation scheme [8], a common and widely used method for BLDC motor control. PWM signals drive the upper MOSFETs of phases A, B, and C to regulate the voltage across the motor windings and control motor speed. Meanwhile, constant turn ON and OFF signals drive the lower MOSFETs to ensure proper commutation of the motor phases.

By meticulously designing the data collection to cover a broad range of motor speeds, torque levels, and commutation conditions, we aim to create a comprehensive and diverse dataset reflecting the complex dynamics of BLDC motor drives. This enriched dataset will facilitate the training and evaluation of deep learning models, enabling us to draw meaningful conclusions about Hall sensor displacement under various operating conditions.

RECORDS AND STORAGE

BLDC-HSD is organized in an Excel file (.xlsx) and consists of measurements classified into four distinct labels based on the delay caused by Hall sensor displacement: *no delay*, *0.0001 delay*, *0.05 delay*, and *0.01 delay*. Each label corresponds to a separate dataset containing 60 000 data points. These measurements are captured at precise intervals of 5×10^{-7} s, resulting in a time span from 0 to 0.03 s for each dataset.

The Excel file is structured with four individual sheets, each representing one of the delay labels. Within each sheet, there are four columns: *time (in seconds)*, *phase A current*, *phase B current*, and *phase C current*. The Time column initiates at 0 s and increments by 5×10^{-7} s for every subsequent row, continuing up to 0.03 s across 60 000 rows. This ensures a detailed temporal resolution for analyzing the data.

The phases A, B, and C current columns contain floating-point numbers that represent the current values in three different phases (A, B, and C) of an electrical system at each time point. These measurements are crucial for understanding the behavior of the electrical system under the

influence of different delay conditions. The current values are captured with high precision, allowing for accurate and detailed analysis.

For example, in each sheet, the data begin with the time column, which starts at 0 s and increments by 5×10^{-7} s for each row. Correspondingly, the phase A current, phase B current, and phase C current columns provide the current measurements at these specific time intervals. This results in a comprehensive set of data for each delay scenario, facilitating a thorough examination of the effects of Hall sensor displacement on the electrical system.

Each of the four sheets in the Excel file offers a unique set of data corresponding to one of the delay labels. The *no delay* sheet contains data where the Hall sensor displacement has no delay effect, serving as a baseline for comparison. The *0.0001 delay*, *0.05 delay*, and *0.01 delay* sheets contain data where the Hall sensor displacement causes slight, moderate, and minimal delays, respectively. This structured approach allows for a detailed comparative analysis of the different delay effects on the phase currents.

In summary, the Excel file is a well-organized repository of data with four sheets, each representing a different delay scenario. Each sheet contains 60 000 rows of data with time intervals of 5×10^{-7} s, and four columns detailing the time and the current measurements for phases A, B, and C. This comprehensive set of data is invaluable for analyzing the impact of Hall sensor displacement delays on an electrical system.

INSIGHTS AND NOTES

This study [1] has demonstrated the significance of accurate fault detection in BLDC motors, particularly in identifying issues caused by Hall sensor displacement. The comprehensive BLDC-HSD dataset was developed for this purpose, encompassing various levels of sensor displacement, has proven instrumental in training and evaluating deep learning models designed for fault detection.

By categorizing phase current measurements into distinct fault conditions and organizing them meticulously, the BLDC-HSD dataset has facilitated detailed analysis and model training. The structured data, captured with high temporal resolution, have enabled the development of robust deep learning models capable of distinguishing between normal and faulty sensor conditions with high accuracy.

The successful application of BLDC-HSD underscores the potential of deep learning approaches in enhancing the reliability and efficiency of BLDC motor systems. Improved fault detection and classification can lead to timely maintenance and reduced risk of motor failure, thereby contributing to overall system stability and performance. Future work will focus on further refining these models and exploring additional data-driven strategies to advance fault diagnosis and motor control technologies.

In general, this BLDC-HSD dataset highlights the critical role of well-organized data in the development of effective fault detection systems and paves the way for further advances in motor fault diagnosis through deep learning.

SOURCE CODE AND SCRIPTS

Python source code used is available on Github: <https://github.com/yklee1014/BLDC-Fault-Detection>.

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All authors wrote and reviewed the manuscript.

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